

PYTHON PROGRAMMING FOR DATA ANALYTICS

Assignment Report

Submitted By: Mayank Gupta

Course: BCA

College: Future University

Python source code, brief explanations, and program-output screenshots for all assigned tasks.

Task 1: Task 1 – Python Basics

Brief Explanation

This program demonstrates basic Python input and output using input(), print(), variables, and formatted strings.

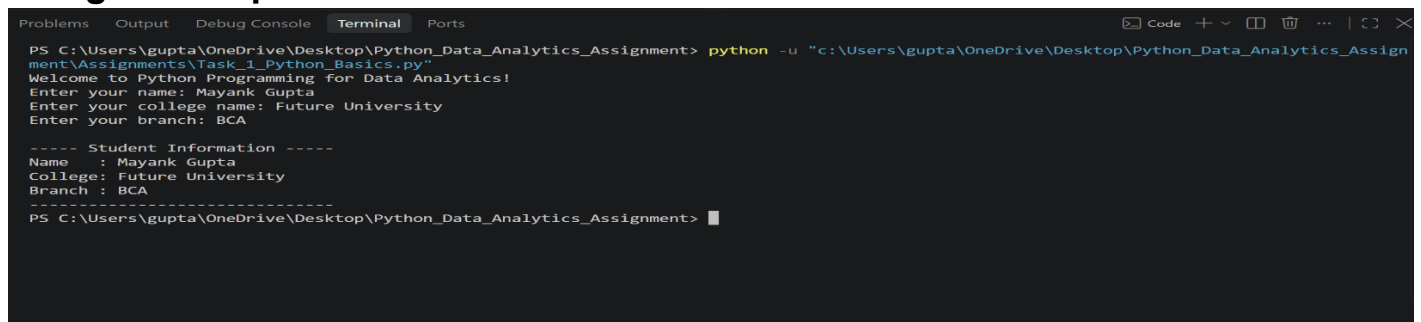
Python Source Code

```
# Task 1: Python Basics
print("Welcome to Python
Programming for Data Analytics!")

name = input("Enter your name: ")
college = input("Enter your college name: ")
branch = input("Enter your branch: ")

print("\n----- Student Information -----")
print(f"Name : {name}")
print(f"College: {college}")
print(f"Branch : {branch}")
print("-----")
```

Program Output Screenshot



```
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment> python -u "c:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment\Assignments\Task_1_Python_Basics.py"
Welcome to Python Programming for Data Analytics!
Enter your name: Mayank Gupta
Enter your college name: Future University
Enter your branch: BCA

----- Student Information -----
Name : Mayank Gupta
College: Future University
Branch : BCA
-----
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment>
```

Task 2: Task 2 – Variables & Data Types

Brief Explanation

This program demonstrates integer, float, string, and boolean variables and uses type() to display their data types.

Python Source Code

```
# Task 2: Variables and Data
Types age = 21 percentage = 85.5
name = "Mayank Gupta" is_student
= True

print("Variable Values and Data Types")
print("-----")
print("Age:", age)
print("Data Type:", type(age))
print("\nPercentage:", percentage)
print("Data Type:", type(percentage))
print("\nName:", name)
print("Data Type:", type(name))
print("\nIs Student:", is_student)
print("Data Type:", type(is_student))
```

Program Output Screenshot

```
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment> python -u "c:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment\Assignments\Task_2_Variables_DataTypes.py"
Variable Values and Data Types
-----
Age: 21
Data Type: <class 'int'>

Percentage: 85.5
Data Type: <class 'float'>

Name: Mayank Gupta
Data Type: <class 'str'>

Is Student: True
Data Type: <class 'bool'>
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment> █
```

Task 3: Task 3 – Operators / Calculator

Brief Explanation

This calculator demonstrates arithmetic operators: addition, subtraction, multiplication, division, and modulus.

Python Source Code

```
# Task 3: Calculator using Operators num1
= float(input("Enter first number: "))
num2 = float(input("Enter second number:
"))

print("\n----- Calculator Results -----")
print("Addition      :", num1 + num2)
print("Subtraction   :", num1 - num2)
print("Multiplication :", num1 * num2)

if num2 != 0:
    print("Division      :", num1 /
num2)    print("Modulus      :", num1
% num2) else:
    print("Division      : Cannot divide by zero")
print("Modulus      : Cannot divide by zero") print("-----
-----")
```

Program Output Screenshot

```
Problems Output Debug Console Terminal Ports
Enter first number: 69
Enter second number: 79

----- Calculator Results -----
Addition      : 148.0
Subtraction   : -10.0
Multiplication : 5451.0
Division      : 0.8734177215189873
Modulus       : 69.0
-----
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment> █
```

Task 4: Task 4 – Conditional Statements

Brief Explanation

This program uses if, elif, and else to determine a grade from the marks entered by the user.

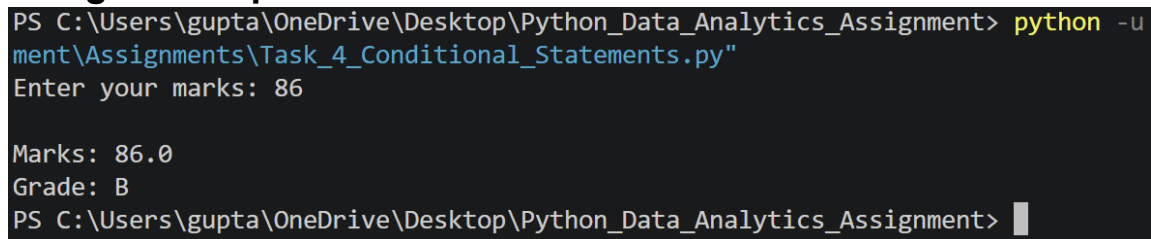
Python Source Code

```
# Task 4: Grade Calculation marks =
float(input("Enter your marks: "))

if marks >= 90:
    grade = "A" elif
marks >= 75:
    grade = "B" elif
marks >= 60:
```

```
grade = "C" else:  
grade = "Fail"  
  
print("\nMarks:", marks)  
print("Grade:", grade)
```

Program Output Screenshot



```
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment> python -u  
ment\Assignments\Task_4_Conditional_Statements.py  
Enter your marks: 86  
  
Marks: 86.0  
Grade: B  
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment> █
```

Task 5: Task 5 – Loops

Brief Explanation

This task demonstrates for and while loops through number printing, multiplication tables, and even-number generation.

Python Source Code

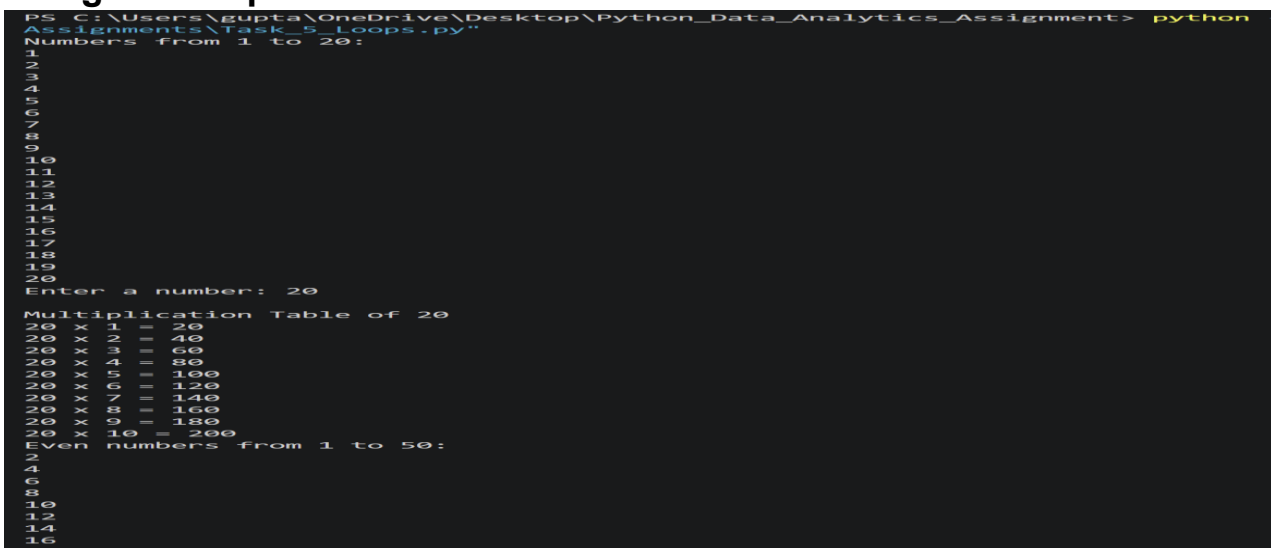
```
# Task 5: Loops

# 5.1 Numbers from 1 to 20
print("Numbers from 1 to 20:") for i in range(1, 21):
    print(i)

# 5.2 Multiplication table number = int(input("\nEnter a number
for multiplication table: ")) print(f"\nMultiplication Table of
{number}") for i in range(1, 11):    print(f"{number} x {i} =
{number * i}")

# 5.3 Even numbers from 1 to 50 using while
loop number = 2 print("\nEven numbers from 1
to 50:") while number <= 50:    print(number)
    number += 2
```

Program Output Screenshot



```
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment> python
Assignments\Task_5_Loops.py
Numbers from 1 to 20:
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Enter a number: 20
Multiplication Table of 20
20 x 1 = 20
20 x 2 = 40
20 x 3 = 60
20 x 4 = 80
20 x 5 = 100
20 x 6 = 120
20 x 7 = 140
20 x 8 = 160
20 x 9 = 180
20 x 10 = 200
Even numbers from 1 to 50:
2
4
6
8
10
12
14
16
```

Task 6: Task 6 – Functions

Brief Explanation

Two user-defined functions are used to calculate the square of a number and the average of three numbers.

Python Source Code

```
# Task 6: User-Defined Functions
def calculate_square(number):
    return number ** 2

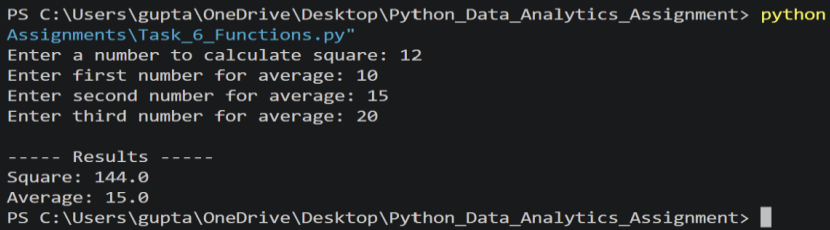
def calculate_average(num1, num2, num3):
    return (num1 + num2 + num3) / 3

number = float(input("Enter a number to calculate square:
")) a = float(input("Enter first number for average: ")) b
= float(input("Enter second number for average: ")) c =
float(input("Enter third number for average: "))

square = calculate_square(number)
average = calculate_average(a, b, c)
```

```
print("\n----- Results -----")
print("Square:", square)
print("Average:", average)
```

Program Output Screenshot



```
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment> python
Assignments\Task_6_Functions.py
Enter a number to calculate square: 12
Enter first number for average: 10
Enter second number for average: 15
Enter third number for average: 20

----- Results -----
Square: 144.0
Average: 15.0
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment> █
```

Task 7: Task 7 – Strings & Collections

Brief Explanation

This program demonstrates string methods, list operations, tuple indexing, dictionary storage, and set operations.

Python Source Code

```
# Task 7: Strings and Collections

# String Operations text = "Python Data Analytics"
print("----- String Operations -----") print("Original
String:", text) print("Uppercase:", text.upper())
print("Lowercase:", text.lower()) print("Replace:",
text.replace("Python", "Advanced Python")) print("Find
'Data':", text.find("Data"))

# List Operations print("\n----- List
Operations -----") fruits =
["Apple", "Banana", "Mango"]
print("Original List:", fruits)
fruits.append("Orange") print("After
append:", fruits)
fruits.remove("Banana") print("After
remove:", fruits) fruits.sort()
print("After sort:", fruits)

# Tuple print("\n----- Tuple -----")
student_tuple = ("Mayank", 21, "BCA")
print("Tuple:", student_tuple)
print("First Element:", student_tuple[0])
print("Second Element:",
student_tuple[1])

# Dictionary print("\n-----
Dictionary -----") student = {
"name": "Mayank Gupta",
"age": 21,
"branch": "BCA",
"college": "Future
University" } print("Student
Information:") for key, value in
student.items(): print(key,
":", value)

# Set print("\n----- Set Operations
-----") numbers = {10, 20, 30}
print("Original Set:", numbers)
numbers.add(40) print("After add:",
numbers) numbers.remove(20)
print("After remove:", numbers)
```

Program Output Screenshot

```
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment> python -
Assignments\Task_7_Strings_Collections.py"
----- String Operations -----
Original String: Python Data Analytics
Uppercase: PYTHON DATA ANALYTICS
Lowercase: python data analytics
Replace: Advanced Python Data Analytics
Find 'Data': 7

----- List Operations -----
Original List: ['Apple', 'Banana', 'Mango']
After append: ['Apple', 'Banana', 'Mango', 'Orange']
After remove: ['Apple', 'Mango', 'Orange']
After sort: ['Apple', 'Mango', 'Orange']

----- Tuple -----
Tuple: ('Mayank', 21, 'BCA')
First Element: Mayank
Second Element: 21

----- Dictionary -----
Student Information:
name : Mayank Gupta
age : 21
branch : BCA
college : Future University

----- Set Operations -----
Original Set: {10, 20, 30}
After add: {40, 10, 20, 30}
After remove: {40, 10, 30}
PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment>
```

Task 8: Task 8 – Basic File Handling

Brief Explanation

This program creates a text file, writes an introduction, reads the file, and displays its contents.

Python Source Code

```
# Task 8: Basic File Handling
filename = "introduction.txt"

introduction = """ My
name is Mayank Gupta.
I am pursuing BCA from Future University.
I am interested in Data Analytics.
I am learning Python, SQL, Excel, and Power BI.
"""

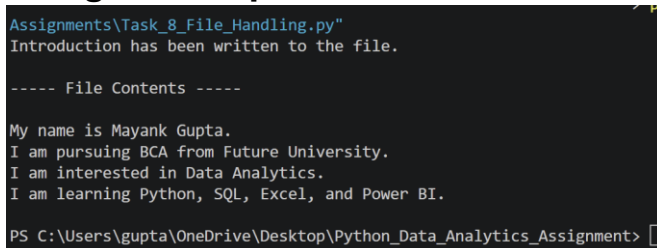
with open(filename, "w") as file:
    file.write(introduction)

print("Introduction has been written to the file.")

with open(filename, "r") as file:
    content = file.read()

print("\n----- File Contents -----")
print(content)
```

Program Output Screenshot



```
Assignments\Task_8_File_Handling.py"
Introduction has been written to the file.

----- File Contents -----

My name is Mayank Gupta.
I am pursuing BCA from Future University.
I am interested in Data Analytics.
I am learning Python, SQL, Excel, and Power BI.

PS C:\Users\gupta\OneDrive\Desktop\Python_Data_Analytics_Assignment>
```

Task 9: Task 9 – Student Record Management System

Brief Explanation

This mini-project stores student dictionaries in a list and provides add, display, search, delete, and exit operations.

Python Source Code

```
# Task 9: Student Record Management
System students = []

def add_student():
    print("\n----- Add Student -----")
    name = input("Enter student name: ")
    age = int(input("Enter student age: "))
    branch = input("Enter branch: ")
    college = input("Enter college name: ")

    student = {
        "name": name,
        "age": age,
        "branch": branch,
        "college": college
    }
    students.append(student)
    print("\nStudent record added successfully!")

def display_students():
    print("\n----- All Student Records -----")
    if len(students) == 0:
        print("No student records found.")
        return
```

```

        for index, student in enumerate(students, start=1):
            print(f"\nStudent {index}")
            print("Name      :", student["name"])
            print("Age       :", student["age"])
            print("Branch    :", student["branch"])
            print("College:", student["college"])

def search_student():
    print("\n----- Search Student -----")
    search_name = input("Enter student name to search: ")
    found = False

    for student in students:
        if student["name"].lower() == search_name.lower():
            print("\nStudent Found!")
            print("Name      :", student["name"])
            print("Age       :", student["age"])
            print("Branch    :", student["branch"])
            print("College:", student["college"])
            found = True
            break

    if not found:
        print("Student record not found.")

def delete_student():
    print("\n----- Delete Student -----")
    delete_name = input("Enter student name to delete: ")

    for student in students:
        if student["name"].lower() == delete_name.lower():
            students.remove(student)
            print("Student record deleted successfully!")
            return

    print("Student record not found.")

while True:

    print("\n=====")
    print(" Student Record Management")
    print("=====")
    print("1. Add Student")      print("2. Display All Students")
    print("3. Search Student")  print("4. Delete Student")
    print("5. Exit")
    print("=====")
    choice = input("Enter your choice: ")

    if choice == "1":
        add_student()
    elif choice == "2":
        display_students()
    elif choice == "3":
        search_student()
    elif choice == "4":
        delete_student()
    elif choice == "5":
        print("Thank you for using the Student Record Management System.")
        break
    else:
        print("Invalid choice. Please try again.")

```

Program Output Screenshot

```
Student Record Management
=====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Exit
=====
Enter your choice: 1

----- Add Student -----
Enter student name: Mayank Gupta
Enter student age: 20
Enter branch: BCA
Enter college name: FU

Student record added successfully!

Student Record Management
=====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Exit
=====
Enter your choice: 2

----- All Student Records -----

Student 1
Name : Mayank Gupta
Age : 20
Branch : BCA
College: FU

Student Record Management
=====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Exit
```

Conclusion

This assignment builds a practical foundation in Python programming for Data Analytics. It covers input/output, data types, operators, conditional statements, loops, functions, strings, collections, file handling, and a mini Student Record Management System.

Google Drive Link :

https://drive.google.com/drive/folders/1Glr_a1Rt3xzOOTBMm_79Fqx67Or2h0AY?usp=drive_link